



Service Tool Instruction

Service Tool Instruction
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Description

Engine-Lifting Fixture

Purpose

This service tool instruction details the use of engine-lifting fixture, Part Number 3163264, for the K19, K38, K50, QSK19, QST30, QSK38, QSK45, QSK50, QSK60, and QSK78 engines.

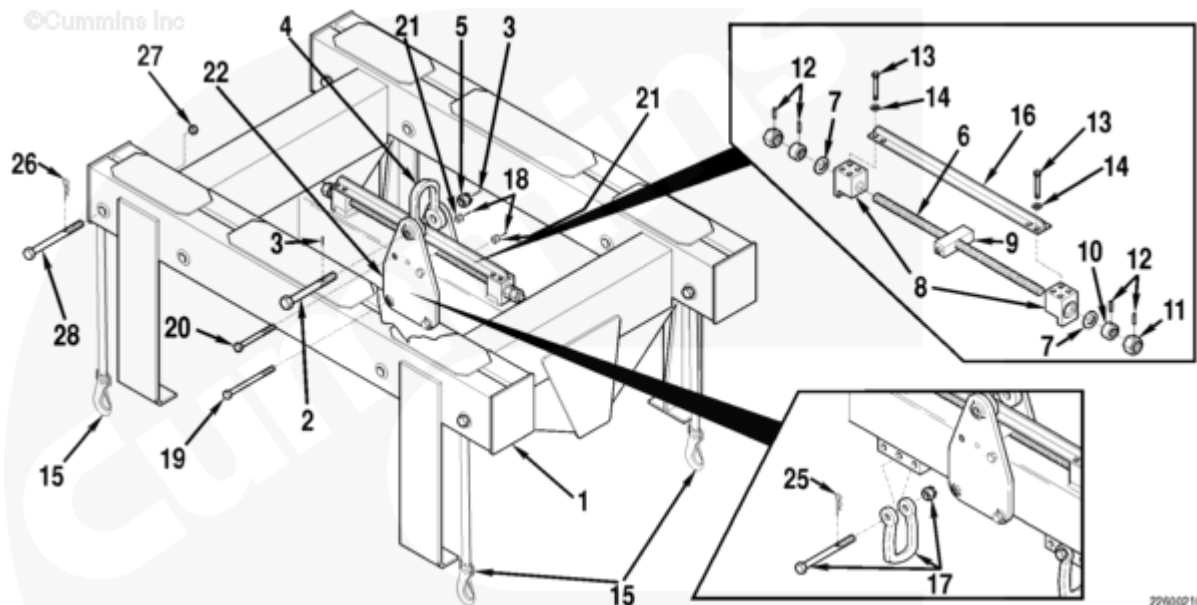


Table 1, Engine-
Lifting Fixture, Part
Number 3163264

Item Number	Part Number	Description	Quantity
1	3163265	Engine-lifting fixture	1
2	3163272	Capscrew	1
3	3163873	Cotter pin	1
4	2892357	Lifting clevis	1
5	3163848	Nut	1
6	3164515	Adjusting screw	1
7	3164523	Plain washer	2

8	3164521	Adjusting block	2
9	3164518	Adjusting block	1
10	3164517	Adjusting nut	2
11	3164516	Adjusting nut	2
12	3164552	Roll pin	4
13	3164519	Capscrew	8
14	3164522	Plain washer	8
15	3163267	Lifting sling	4
16	3164524	Plate	1
17	3163266	Shackle, castle nut, and capscrew	2
18	3163872	Cotter pin	4
19	3163273	Capscrew	2
20	3164582	Capscrew	2
21	3163847	Slotted nut	4
22	3164520	Plate	2
Not shown	3163851	Wear plate	2
Not shown	3163850	Spacer	2
25	3163875	Hairpin	2
26	3163876	Hairpin	4
27	S-215	Nut	4
28	3163274	Capscrew	4

Safety Guidelines:

WARNING

This engine-lifting fixture must be used for lifting the specified engines only. Improper use can result in equipment damage, severe personal injury, or death.

WARNING

Rigging and lifting must be done by trained and experienced personnel only. Improper use can result in equipment damage, severe personal injury, or death.

WARNING

Personnel must stay clear of the engine and engine-lifting fixture while the engine is lifted. Failure to do so can result in severe personal injury or death.

WARNING

The adjusting screw must not be adjusted while the engine-lifting fixture is loaded. Doing so can result in equipment damage, severe personal injury, or death.

WARNING

The engine must not swing while it is lifted with the engine-lifting fixture. Doing so can result in equipment damage, severe personal injury, or death.

The following safety instructions are in accordance with the Occupational Safety and Health Administration's (OSHA) regulations for rigging equipment for handling material, Standard Number 1926.251.

General

- The engine-lifting fixture (1) and **all** of its components will be inspected prior to each use and as necessary during its use. Damaged or defective equipment **must** be removed from service.
- Store the engine-lifting fixture and its components in a safe place and away from the immediate work area when **not** in use.
- Hooks, links, makeshift fasteners (formed from bolts, rods, and so forth), or other such attachments **must not** be used.
- A thorough inspection **must** be made on a regular basis, based on the frequency of sling use, severity of service conditions, nature of the lifts being made, and experience gained on the service life of slings used in similar circumstances. Inspection intervals **must not** be longer than 12 months.
- The shop should maintain a record of the most recent month in which each sling assembly was thoroughly inspected and make the record available for inspection.
- Operators **must** be trained and experienced in hoisting objects.

Slings:

- Inspection of the slings for the following conditions **must** be done before each use.
- Cuts to the sling cover exposing red-striped white-core yarns
- Holes, tears, snags, or abrasions exposing red-striped white-core yarns
- End fittings that are pitted, corroded, cracked, distorted, or broken
- The sling shows signs of melting, charring, or chemical damage
- Capacity tag is illegible or missing
- Other visible damage that causes doubt as to the strength of the sling.

If any of the above conditions are observed, the sling **must** be removed from service.

It is recommended to store slings in a safe place when **not** in use.

Environmental Conditions

- Nylon and polyester are seriously degraded at temperatures above 93°C [199°F].

- Many chemicals have an adverse effect on nylon and polyester. See Table 2, General Guide for Chemical Effects on Nylon and Polyester.

Table 2, General Guide for Chemical Effects on Nylon and Polyester		
Chemical	Nylon	Polyester
Acids	Not OK	Disintegrated by concentrated sulfuric acid
Alcohols	OK	OK
Aldehydes	OK	Not OK
Strong Alkalis	OK	Degraded by strong alkalis at elevated temperatures
Bleaching Agents	Not OK	OK
Dry Cleaning Solvents	OK	OK
Ethers	OK	Not OK
Halogenated Hydrocarbons	OK	OK
Ketones	OK	OK
Oils, Crude	OK	OK
Oils, Lubricating	OK	OK
Soap and Detergents	OK	OK
Water and Sea Water	OK	OK
Weak Alkalis	OK	OK

Note : Not all safety considerations or situations can be covered in this instruction; so if a situation, item, procedure, or even a step within the procedure seems unsafe, stop and notify your immediate supervisor. If further information is necessary, contact Cummins Inc. at 1-800-CUMMINS (1-800-286-6467) toll free in the U.S. and Canada.